

Preprocessing Steps

Electric Vehicle Charge and Range Analysis — Data Preparation

1. Data Collection

- Imported the Electric Vehicle dataset into MySQL and connected it to Tableau.

2. Data Cleaning

- Removed duplicate records.
- Checked for and handled missing (NULL) values.
- Corrected inconsistent and invalid data entries.

3. Split the “cheapestelectriccars-evdatabase+” Column

- Split the “cheapestelectriccars-evdatabase+” column into three separate columns using Tableau's Split feature to extract meaningful information.

4. Rename the Split Column

- Renamed Split 3 to Brands for better readability and easier analysis.

5. Change Data Type

- Changed the Address field data type to Geographic Role → Address to enable map-based visualizations in Tableau.

6. Data Type Validation

- Verified and corrected data types for text, numeric, and categorical fields.

7. Feature Selection

- Selected relevant fields such as Brand, Vehicle Name, Price, Battery Capacity, Range, Charging Type, Region, and Address.

8. Calculated Fields

- Created calculated fields for aggregations such as charging station count and brand-wise analysis.

9. Data Extraction

- Converted the MySQL Live connection to a Tableau Extract to improve performance and support publishing to Tableau Public.

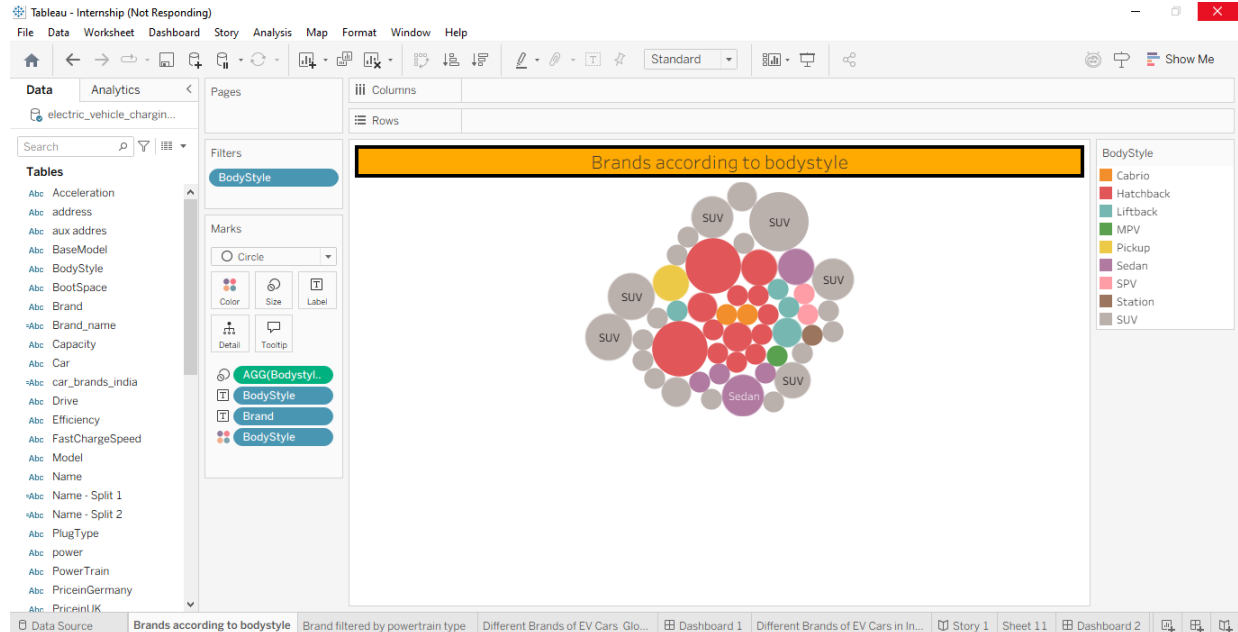
10. Data Validation

- Verified the processed data by checking row counts, field values, and visualizations before creating dashboards.

Business Questions with Visualizations

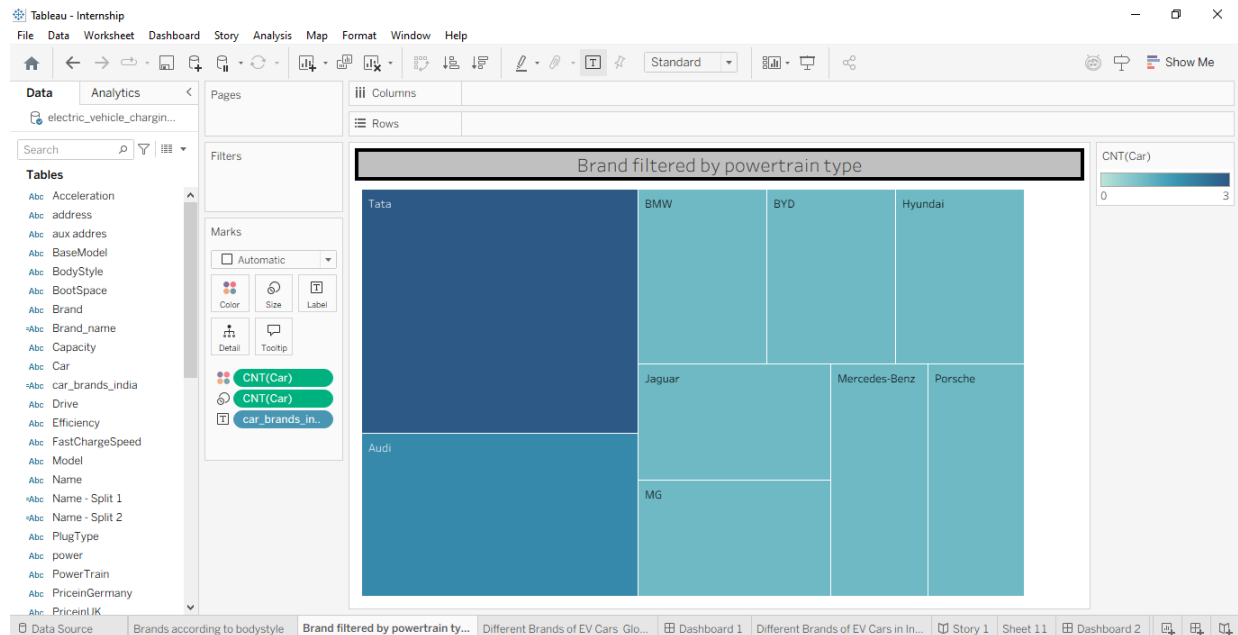
Electric Vehicle Charge and Range Analysis — Tableau Dashboards

Q1. Which EV brands offer which body styles?



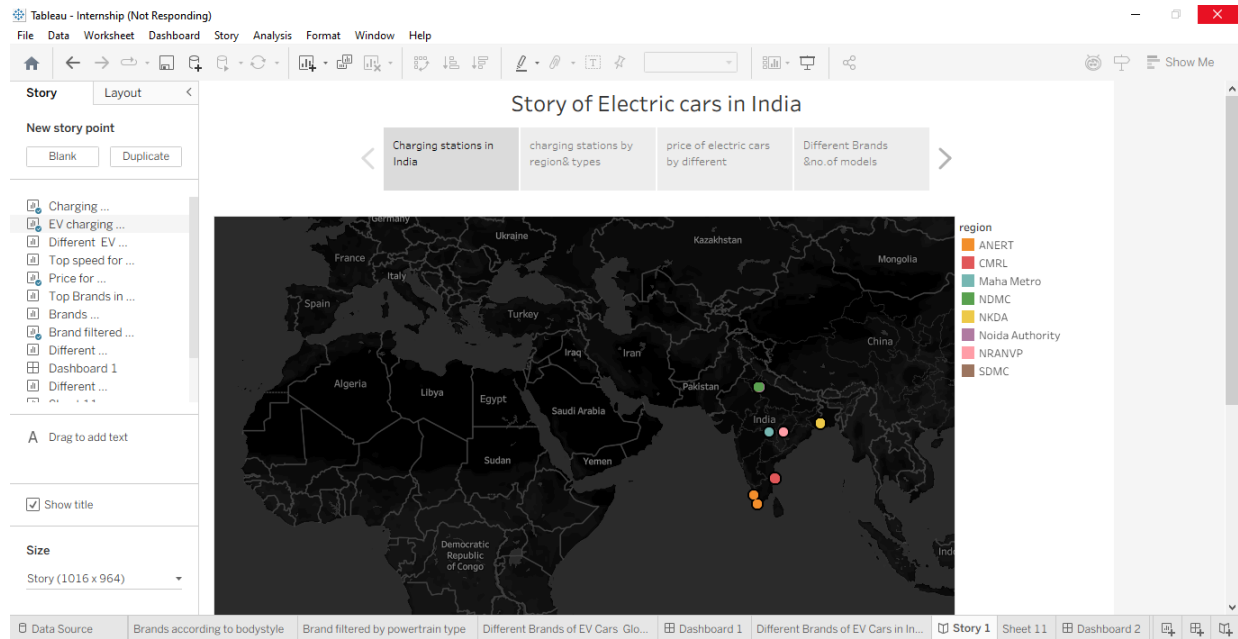
Brands According to Bodystyle

Q2. How are EV brands distributed across different powertrain types?



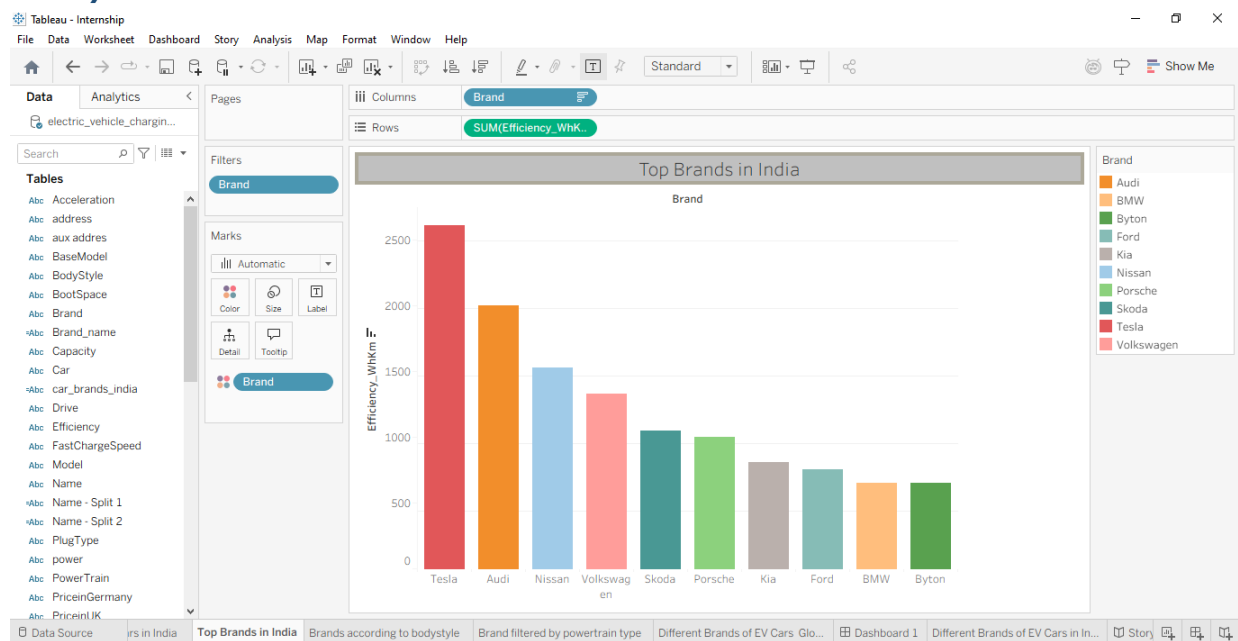
Brand Filtered by Powertrain Type

Q3. What is the overall story of EV charging infrastructure, pricing, and brand availability in India?



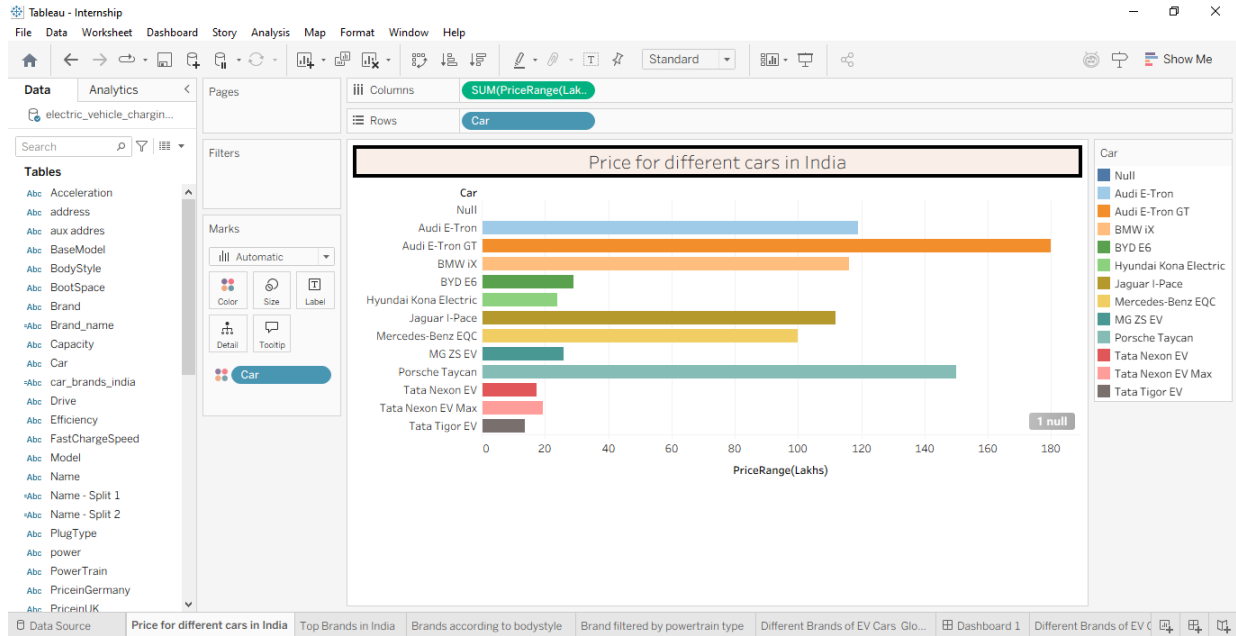
Story of Electric Cars in India

Q4. Which are the top-performing EV brands in India based on efficiency (Wh/Km)?



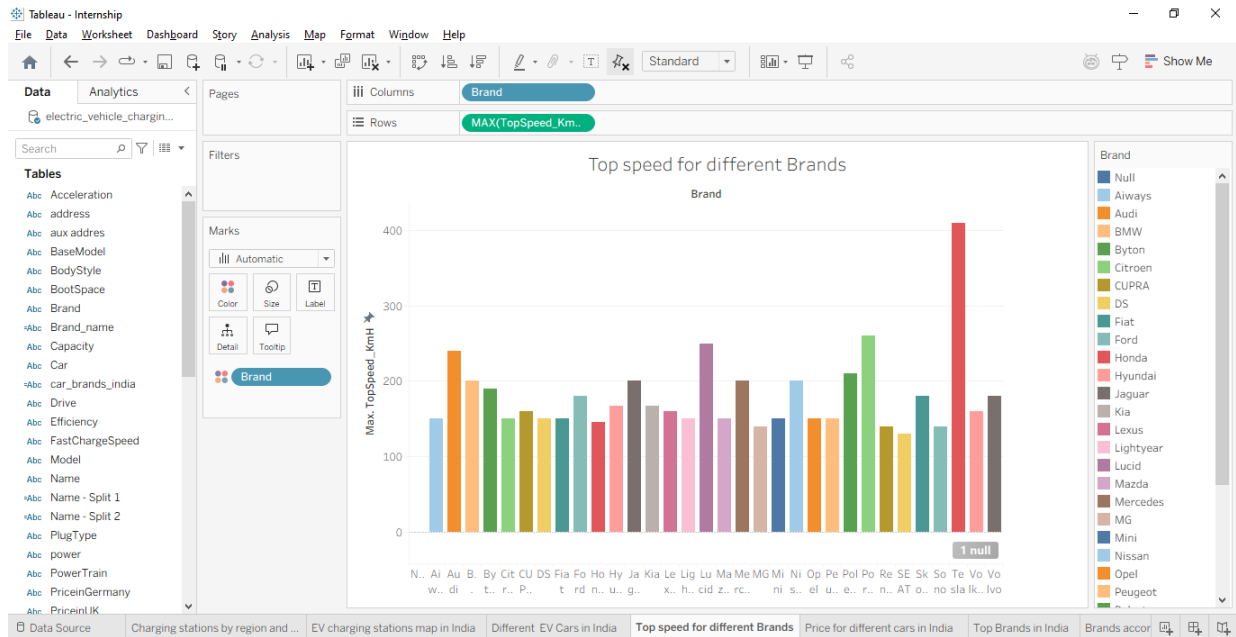
Top Brands in India

Q5. What is the price range of different EV car models available in India?



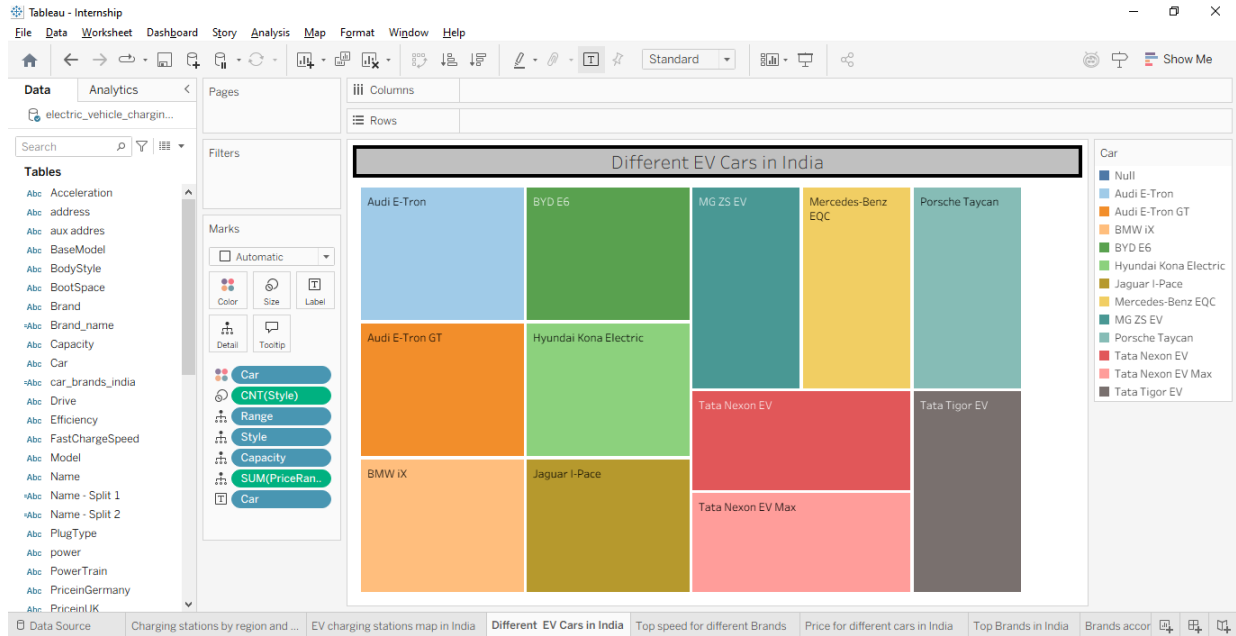
Price for Different Cars in India

Q6. Which EV brands achieve the highest top speed?



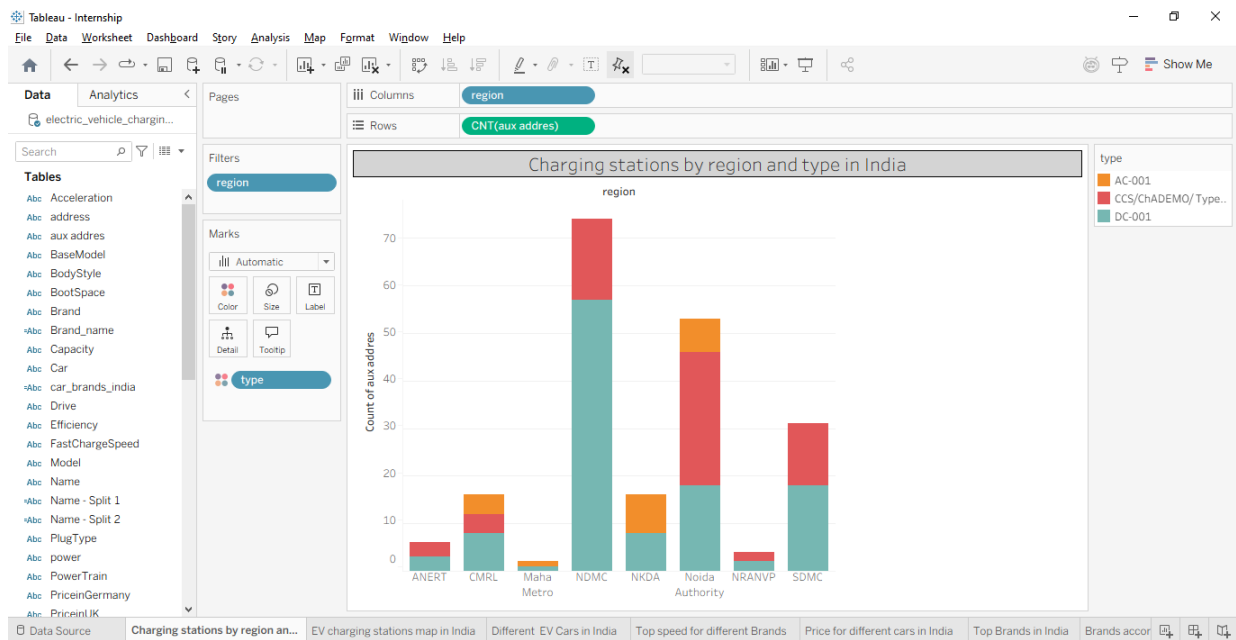
Top Speed for Different Brands

Q7. What different EV car models are currently available in the Indian market?



Different EV Cars in India

Q8. How are EV charging stations distributed by region and connector type in India?



Charging Stations by Region and Type in India